

Laboratory Exercise: Phylogenetic Analysis Using NCBI and Galaxy

Course: Cell and Molecular Biology Laboratory

Activity: Group work (4–5 students per group)

Date: July 31, 2026

Group Number: 02

Taxonomic Group:

Frogs

Names of Group Members:

1. ESTRELLANES, KLEA MAE G.
2. VILLEGAS, KYLA ROSE D.
3. GALVEZ, WILLHELM WIEN L.

Introduction

Phylogenetic analysis is used to examine the evolutionary relationships among organisms using molecular sequence data. In this laboratory exercise, mitochondrial cytochrome c oxidase subunit I (COI/COX1) sequences from selected frog species were retrieved from the NCBI Nucleotide database. The sequences were analyzed in Galaxy through multiple sequence alignment, consensus sequence generation, and phylogenetic tree construction to examine the evolutionary relationships among the selected taxa.

Biological Question

How are the selected frog species evolutionarily related based on their mitochondrial cytochrome c oxidase subunit I (COI/COX1) gene sequences?

The selected frog species are evolutionarily related according to similarities and differences in their mitochondrial COI gene sequences. The phylogenetic tree shows that species with more similar COI sequences cluster together into closely related groups. *Aquarana catesbeiana* and *Lithobates clamitans*, *Pelophylax ridibundus* and *Pelophylax lessonae*, and *Hyla arborea* and *Dryophytes cinereus* each form sister taxa, indicating that they share more recent common

ancestors. The three *Rana* species share a common ancestry but diverged sequentially rather than forming a single sister pair. *Xenopus laevis*, selected as the outgroup, represents a more distantly related lineage and serves as a reference for interpreting the evolutionary relationships among the other frog species. Overall, the COI gene sequences indicate that closely related species cluster together according to their evolutionary history.

Table of Species, NCBI Accession Numbers, Sequence Lengths, and Outgroup

Scientific name	Common name	NCBI accession	Length (bp)	Focal taxon / Outgroup
<i>Rana temporaria</i>	Common frog	MN993213.1	658	Focal
<i>Rana arvalis</i>	Moor frog	MN993202.1	658	Focal
<i>Rana dalmatina</i>	Agile frog	MN993208.1	658	Focal
<i>Aquarana catesbeiana</i>	American bullfrog	KY587197.1	658	Focal
<i>Lithobates clamitans</i>	Bronze frog	MN135616.1	658	Focal
<i>Pelophylax ridibundus</i>	Marsh frog	MN993187.1	658	Focal
<i>Pelophylax lessonae</i>	Pool frog	MN993185.1	658	Focal
<i>Hyla arborea</i>	European tree frog	MN993119.1	658	Focal
<i>Dryophytes cinereus</i> (syn. <i>Hyla cinerea</i>)	Green tree frog	KT388402.1	658	Focal
<i>Xenopus laevis</i>	African clawed frog	GQ862287.1	652	Outgroup

Methods

NCBI Retrieval

Mitochondrial cytochrome c oxidase subunit I (COI/COX1) sequences were retrieved from the NCBI Nucleotide database. Each frog species was searched individually, and the scientific name, accession number, sequence length, and record title were verified before the sequences were downloaded in FASTA format.

Galaxy

The downloaded FASTA sequences were uploaded to Galaxy (<https://usegalaxy.org>). The datasets were verified as FASTA files, renamed when necessary, and organized in a Galaxy History for subsequent analyses.

MUSCLE Alignment

The individual COI sequences were combined into a single multi-FASTA dataset and aligned using the MUSCLE multiple sequence alignment tool in Galaxy with the default settings. The alignment was inspected before being saved for phylogenetic analysis.

Consensus Generation

The MUSCLE alignment was used as input to generate a consensus sequence using the available consensus sequence tool in Galaxy. The consensus sequence was examined together with the conserved and variable nucleotide sites.

Tree Construction

The final MUSCLE nucleotide alignment was used as input for FastTree in Galaxy to construct a phylogenetic tree. The resulting tree was visualized using a compatible tree viewer, with the selected outgroup used to root and interpret the evolutionary relationships among the focal taxa.

Representative MUSCLE Alignment Figure/Screenshot

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MN993213.1 Rana temporaria voucher Herp209 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
MN993202.1 Rana arvalis voucher Herp 88 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
MN993208.1 Rana dalmatina voucher Herp 83 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
KY587197.1 Aquarana catesbeiana isolate LiCa_mt cytochrome oxidase subunit I (COI) gene, partial cds; mitochondrial
MN135616.1 Lithobates clamitans voucher USNM:Herp:575057 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
MN993187.1 Pelophylax ridibundus voucher Herp 122 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
MN993185.1 Pelophylax lessonae voucher Herp 31 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
MN993119.1 Hyla arborea voucher Herp193 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
KT388402.1 Hyla cinerea voucher USMS:ADNARL-28 cytochrome oxidase subunit I (COI) gene, partial cds; mitochondrial
GQ862287.1 Xenopus laevis voucher BJE00263 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial
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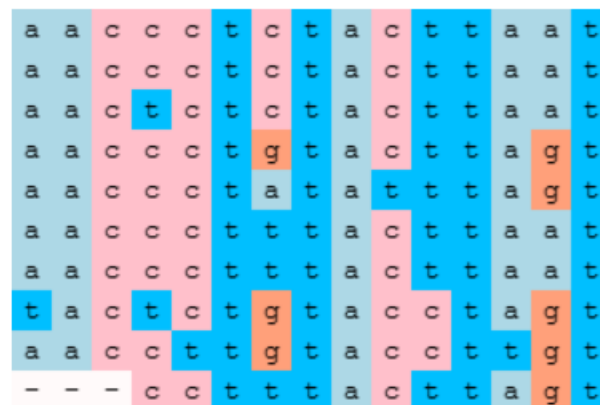


Figure 1. The representative MUSCLE alignment shows that the mitochondrial COI sequences from the selected frog species were successfully aligned so that homologous nucleotide positions occur in the same columns. Most positions in the alignment are conserved across the sequences, indicating that the COI gene is highly similar among the selected taxa.

Several nucleotide positions show differences among the species, as indicated by the different colored bases in the alignment. These variable sites represent nucleotide substitutions that distinguish one species from another and provide the genetic variation needed for phylogenetic analysis.

The alignment also shows a small number of gaps (hyphens), which represent insertion or deletion (indel) events introduced during the alignment to maximize sequence similarity among homologous regions. In the representative alignment, gaps are minimal, suggesting that the selected COI sequences are of comparable length and correspond to the same homologous region.

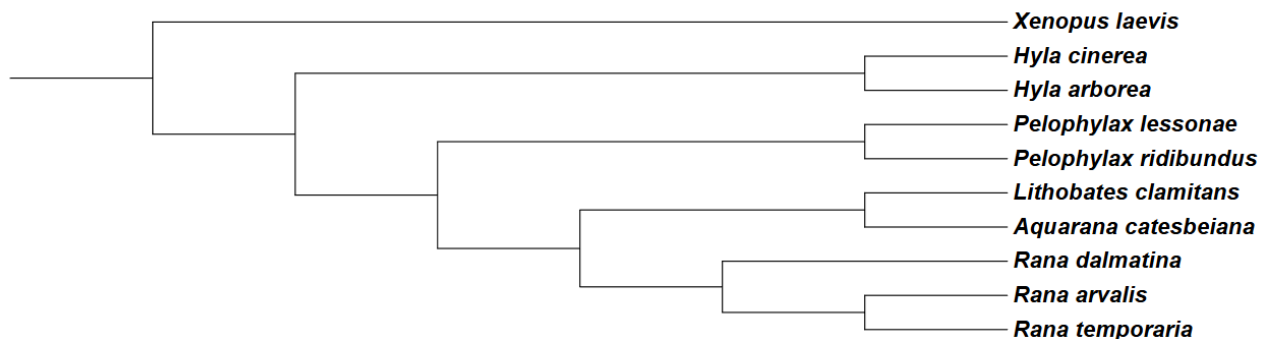
Consensus Sequence or Summary of Conserved and Variable Sites

The consensus sequence was generated from the aligned mitochondrial COI sequences using the consensus sequence tool in Galaxy.

The multiple sequence alignment showed that the COI sequences contained both conserved and variable nucleotide sites. Conserved sites were shared among most or all frog species, whereas variable sites differed among taxa and provided information useful for phylogenetic analysis.

The consensus sequence represents the most common nucleotide present at each position across the aligned COI sequences.

Clearly Labeled Phylogenetic Tree



Interpretation of Clades, Sister Taxa, Branch Lengths, and Outgroup

The phylogenetic tree reconstructed from the mitochondrial cytochrome c oxidase subunit I (COI) gene sequences grouped the selected frog species into several distinct clades, reflecting their inferred evolutionary relationships. Species that share a more recent branching point are interpreted to have a more recent common ancestor than species that diverged earlier in the tree. The branching pattern indicates that the sampled frog species have undergone evolutionary diversification while retaining recognizable groups of closely related taxa.

Several well-defined sister-taxon relationships are evident in the phylogenetic tree. *Aquarana catesbeiana* (KY587197.1) and *Lithobates clamitans* (MN135616.1) form a sister pair, indicating that they share a recent common ancestor before joining the remaining frog taxa. Similarly, *Pelophylax ridibundus* (MN993187.1) and *Pelophylax lessonae* (MN993185.1) cluster together as sister taxa, reflecting their close evolutionary relationship within the genus *Pelophylax*. Another sister pair consists of *Hyla arborea* (MN993119.1) and *Dryophytes cinereus* (KT388402.1), which share an immediate common ancestor and exhibit close genetic similarity based on the analyzed COI sequences.

Within the genus *Rana*, *Rana temporaria* (MN993213.1), *Rana arvalis* (MN993202.1), and *Rana dalmatina* (MN993208.1) occur along the same lineage but do not form a single immediate sister-taxon group. Instead, the branching pattern indicates successive divergence among these species, suggesting that they share a common ancestry while representing distinct evolutionary lineages within the genus.

Xenopus laevis (GQ862287.1) was selected as the outgroup because it belongs to the family Pipidae, a lineage that diverged earlier than the families represented by the focal frog taxa. The inclusion of *X. laevis* provides an external reference for rooting the phylogenetic tree and interpreting the evolutionary relationships among the ingroup species. However, because the

phylogenetic tree generated by FastTree is displayed in an unrooted configuration, *X. laevis* appears as part of one branch of the tree rather than at its base. Re-rooting the tree using *X. laevis* would place it outside the ingroup and provide a clearer representation of the direction of evolutionary divergence.

The branch lengths in the phylogenetic tree represent the amount of genetic change inferred from the mitochondrial COI sequences. Shorter branches indicate fewer accumulated nucleotide substitutions and therefore closer genetic relationships, whereas longer branches reflect greater genetic divergence over evolutionary time. Overall, the branching pattern and branch lengths support the existence of several closely related species pairs while illustrating the broader evolutionary diversification of the selected frog species based on mitochondrial COI gene sequences.

Screenshots of the Main Steps used:

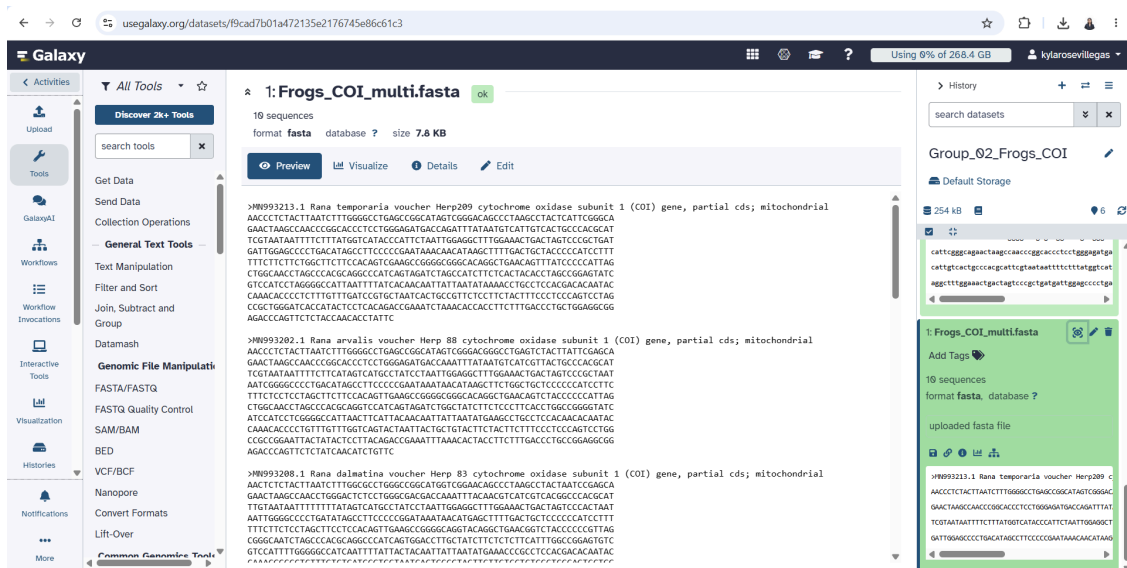


Figure 1. Original multi-FASTA dataset containing the mitochondrial cytochrome *c* oxidase subunit I (COI/COX1) gene sequences of the ten selected frog species uploaded to Galaxy for phylogenetic analysis.

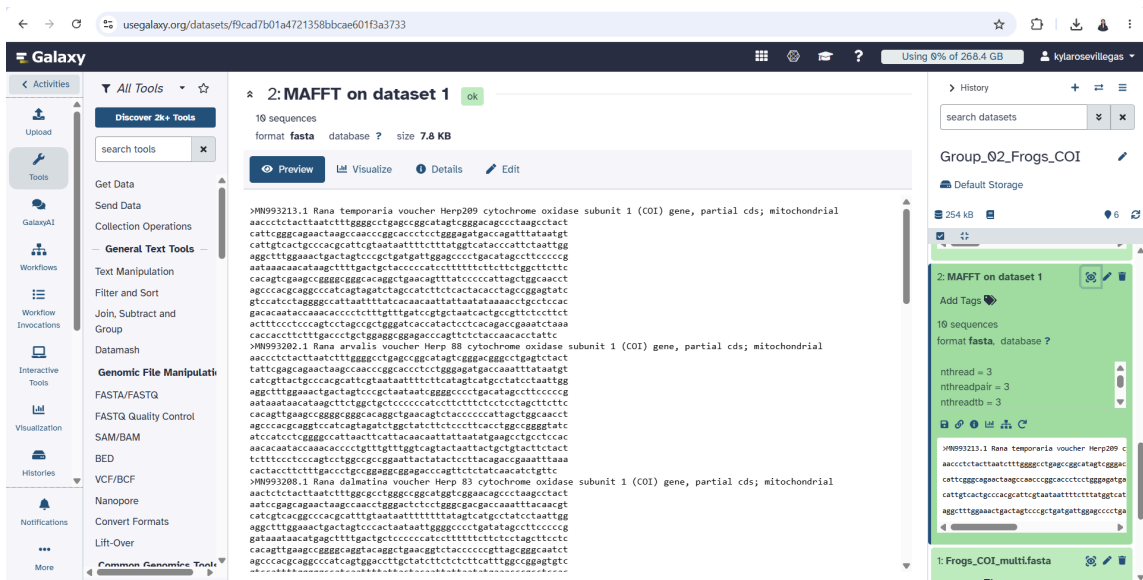
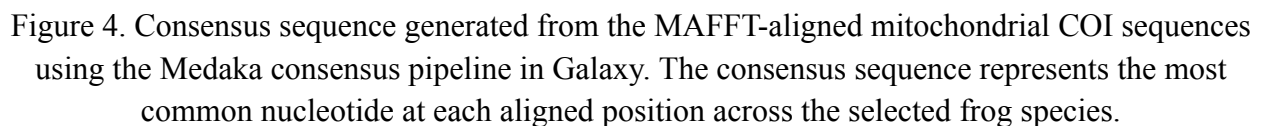
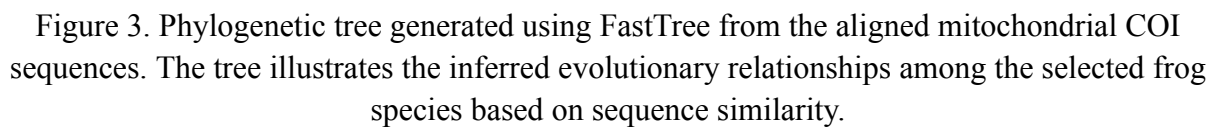


Figure 2. Multiple sequence alignment of the mitochondrial COI gene sequences generated using MAFFT in Galaxy. The alignment arranges homologous nucleotide positions to identify conserved and variable sites among the selected frog species.



Comparison with at Least Two Peer-Reviewed Published Accounts

Paper 1

Citation:

Pyron, R. A., & Wiens, J. J. (2011). A large-scale phylogeny of Amphibia including over 2,800 species, and a revised classification of extant frogs, salamanders, and caecilians. *Molecular Phylogenetics and Evolution*, 61(2), 543–583. <https://doi.org/10.1016/j.ympev.2011.06.012>

Comparison:

The phylogenetic tree generated in this laboratory is generally consistent with the findings of Pyron and Wiens (2011), who reported that closely related frog species tend to cluster according to their evolutionary history inferred from molecular data. In both analyses, species belonging to the same or closely related genera, such as *Pelophylax*, *Lithobates*, and *Hyla/Dryophytes*, form distinct evolutionary groups. Although the present study includes only a limited number of species and uses a single mitochondrial marker (COI), the observed clustering agrees with the broader phylogenetic relationships reported by Pyron and Wiens. Minor differences in branching order may be attributed to the smaller taxon sampling and the use of only one gene sequence in the present analysis.

Paper 2

Citation:

Frost, D. R., Grant, T., Faivovich, J., Bain, R. H., Haas, A., Haddad, C. F. B., de Sá, R. O., Channing, A., Wilkinson, M., Donnellan, S. C., Raxworthy, C. J., Campbell, J. A., Blotto, B. L., Moler, P., Drewes, R. C., Nussbaum, R. A., Lynch, J. D., Green, D. M., & Wheeler, W. C. (2006). The amphibian tree of life. *Bulletin of the American Museum of Natural History*, 297, 1–370. [https://doi.org/10.1206/0003-0090\(2006\)297\[0001:TATOL\]2.0.CO;2](https://doi.org/10.1206/0003-0090(2006)297[0001:TATOL]2.0.CO;2)

Comparison:

The laboratory phylogenetic tree also agrees with the classification proposed by Frost et al. (2006), which recognizes major evolutionary lineages among frogs based on molecular evidence. In both the published study and the present analysis, *Xenopus laevis* represents a more distantly related lineage than the other sampled frogs, making it an appropriate outgroup for phylogenetic reconstruction. Furthermore, species within the same genera or closely related genera show greater genetic similarity and tend to cluster together. Differences between the two phylogenies are expected because Frost et al. analyzed multiple genes and a much larger number of species, whereas this laboratory exercise relied only on mitochondrial COI sequences from ten taxa.

Discussion of Agreements, Disagreements, and Limitations of Using COI Alone

The phylogenetic relationships inferred from the mitochondrial cytochrome c oxidase subunit I (COI) gene in this study are generally consistent with previously published molecular phylogenetic studies of frogs. Closely related species, such as *Pelophylax ridibundus* and *Pelophylax lessonae*, *Aquarana catesbeiana* and *Lithobates clamitans*, and *Hyla arborea* and *Dryophytes cinereus*, clustered together in the phylogenetic tree, supporting their close evolutionary relationships. This overall pattern agrees with the classifications proposed by Frost et al. (2006) and the large-scale molecular phylogeny presented by Pyron and Wiens (2011), which showed that frog species generally cluster according to their evolutionary history and taxonomic relationships.

Some differences between the present phylogenetic tree and published studies are expected. The branching order among certain taxa, particularly within the genus *Rana*, differs slightly from more comprehensive phylogenies. These discrepancies are likely due to the limited number of species included in this laboratory exercise, the use of only a single mitochondrial gene, and the fact that the tree was generated using FastTree from an unrooted alignment. In contrast, published phylogenetic studies typically analyze hundreds to thousands of species using multiple mitochondrial and nuclear genes with more computationally intensive phylogenetic methods, resulting in higher-resolution evolutionary trees.

Although the COI gene is widely used for DNA barcoding and species identification because of its high amplification success and ability to distinguish closely related species, it has limitations when used alone for phylogenetic reconstruction. Because COI is a mitochondrial marker inherited only through the maternal lineage, it represents only a single evolutionary history and may not accurately reflect the complete evolutionary relationships among species. Processes such as incomplete lineage sorting, introgression, hybridization, and differences between mitochondrial and nuclear genomes can lead to phylogenetic patterns that differ from the true species tree. Furthermore, relying on a single gene provides fewer informative characters than analyses based on multiple genes or whole-genome data.

Overall, the mitochondrial COI sequences successfully resolved several closely related frog species and provided a reasonable estimate of their evolutionary relationships. However, incorporating additional mitochondrial and nuclear genes, increasing taxon sampling, and using rooted phylogenetic analyses would improve the robustness and accuracy of the inferred phylogeny.

References

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